

AI Business Intelligence & Forecasting System

📌 Problem

Businesses rely on static reports and lack predictive insights for decision-making.

🚀 Solution

Developed an end-to-end automated business intelligence system integrating:

- Data cleaning (Python, PowerQuery)
- Data storage (MySQL)
- Dashboarding (Power BI)
- Forecasting (Prophet)
- AI-driven insights

🛠️ Tech Stack

- MS Excel (Data Validation, Data Accuracy, Power Query)
- Python (Pandas, NumPy, Prophet, OpenAI, Matplotlib)
- MySQL
- Power BI
- Automation (Task Scheduler)

📊 Features

- Automated ETL pipeline
- KPI generation (Sales, Profit, Delay)
- Sales forecasting (90-day prediction)
- Risk analysis using confidence intervals
- AI-based executive summaries

- Fully automated pipeline

📈 Dashboard Highlights

- Forecast Growth %
- Forecast Risk Range
- Regional Performance
- Sales Trends

💡 Business Impact

- Enables proactive decision-making
- Identifies high-growth regions
- Detects operational risks

⚙️ How to Run

1. Run `master\_pipeline.py`
2. Open Power BI file
3. Click Refresh

📸 Screenshots

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